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LABORATORIO 2 - PROBLEMA 1

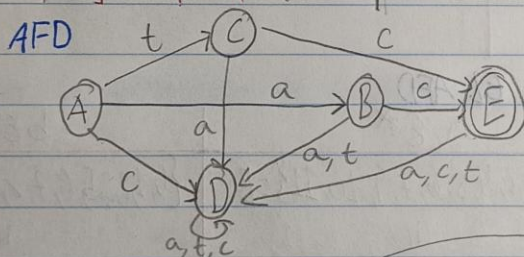
a) $(a|t)c$

TABLA DE TRANSICIONES

	a	t	c	ϵ^*
0	-	-	-	$\{0, 1, 2\}$
1	3	-	-	1
2	-	4	-	2
3	-	-	-	$\{3, 5, 6\}$
4	-	-	-	$\{4, 5, 6\}$
5	-	-	-	$\{5, 6\}$
6	-	-	7	6
7	-	-	-	7

TRANSICIONES PARA EL AFD

ESTADOS	$a\epsilon^*$	$t\epsilon^*$	$c\epsilon^*$
$\{0, 1, 2\} = A$	$\{3, 5, 6\} = B$	$\{4, 5, 6\} = C$	$\{7\} = D$
$\{3, 5, 6\} = B$	$\{7\} = D$	=	$\{7\} = E$
$\{4, 5, 6\} = C$	$\{7\} = D$	=	$\{7\} = E$
$\{5, 6\} = D$	=	=	=
$\{7\} = E$	$\{7\} = D$	=	=



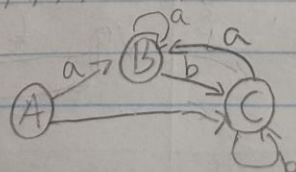
b) $(a|b)^*$

TABLA DE TRANSICIONES

	a	b	ϵ^*
0	-	-	$\{0, 1, 2, 4, 7\}$
1	-	-	$\{1, 2, 4\}$
2	3	-	2
3	-	-	$\{1, 2, 3, 4, 6, 7\}$
4	-	5	4
5	-	-	$\{1, 2, 4, 5, 6, 7\}$
6	-	-	$\{1, 2, 4, 5, 6, 7\}$
7	-	-	7

TRANSICIONES PARA EL AFD

ESTADOS	$a\epsilon^*$	$b\epsilon^*$
$\{0, 1, 2, 4, 7\} = A$	$\{1, 2, 3, 4, 6, 7\} = B$	$\{1, 2, 4, 5, 6, 7\} = C$
$\{1, 2, 3, 4, 6, 7\} = B$		
$\{1, 2, 4, 5, 6, 7\} = C$		

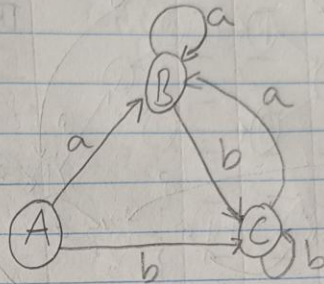
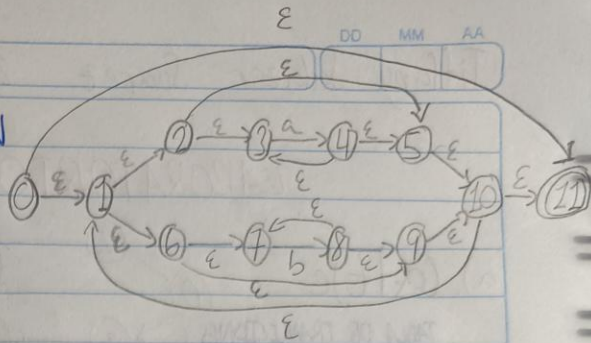


c). $(a^* | b^*)^*$

TABLA DE TRANSICIONES

	a	b	ϵ^*
0	-	-	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$
1	-	-	$\{1, 2, 3, 5, 6, 7, 9, 10, 11\}$
2	-	-	$\{1, 2, 3, 5, 6, 7, 9, 10, 11\}$
3	4	-	3
4	-	-	$\{1, 2, 3, 4, 5, 6, 7, 9, 10, 11\}$
5	-	-	$\{1, 2, 3, 5, 6, 7, 9, 10, 11\}$
6	-	-	$\{1, 2, 3, 5, 6, 7, 9, 10, 11\}$
7	-	8	7
8	-	-	$\{1, 2, 3, 5, 6, 7, 9, 10, 11\}$
9	-	-	$\{1, 2, 3, 5, 6, 7, 9, 10, 11\}$
10	-	-	$\{1, 2, 3, 5, 6, 7, 9, 10, 11\}$
11	-	-	11

AFN



AFD

TABLA DE TRANSICIONES PARA EL AFD

ESTADOS	$a \epsilon^*$	$b \epsilon^*$
$\{0, 1, 2, 3, 5, 6, 7, 9, 10, 11\} = A$	$\{1, 2, 3, 4, 5, 6, 7, 9, 10, 11\} = B$	$\{0, 1, 2, 3, 5, 6, 7, 8, 9, 10, 11\} = C$
$\{1, 2, 3, 4, 5, 6, 7, 9, 10, 11\} = B$	//	//
$\{0, 1, 2, 3, 5, 6, 7, 8, 9, 10, 11\} = C$	//	//

d). $((\epsilon | a) | b^*)^*$

AFN

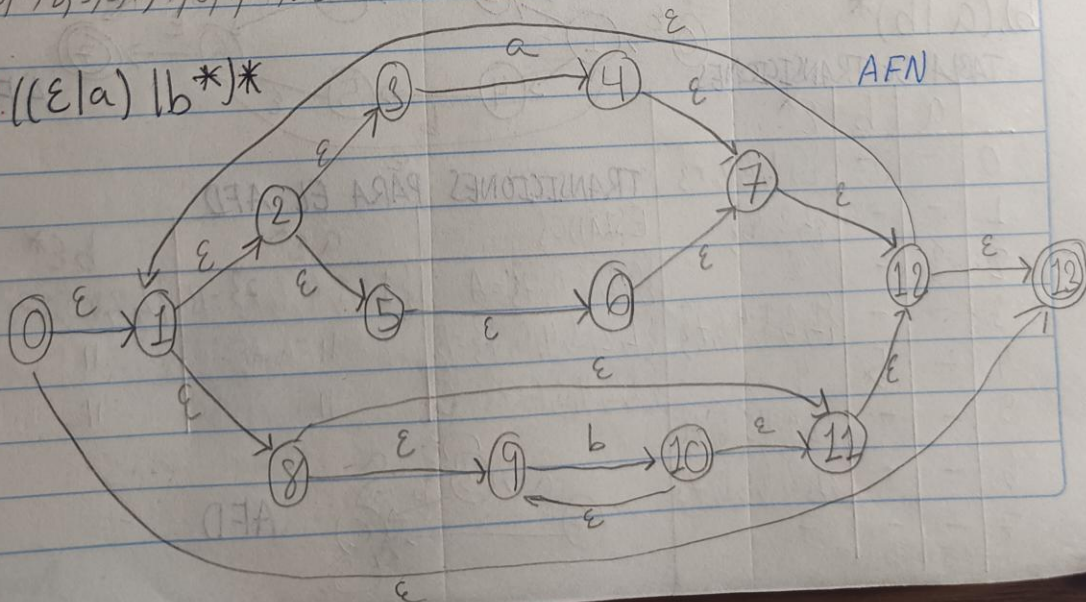


TABLA DE TRANSICIÓN

	a	b	ϵ^*
0	-	-	$\{0,1,2,3,5,6,7,8,9,11,12,13\}$
1	-	-	$\{1,2,3,5,6,7,8,9,11,12,13\}$
2	-	-	$\{1,2,3,5,6,7,8,9,11,12,13\}$
3	4	-	3
4	-	-	$\{1,2,3,4,5,6,7,8,9,11,12,13\}$
5	-	-	$\{1,2,3,5,6,7,8,9,11,12,13\}$
6	-	-	$\{1,2,3,5,6,7,8,9,11,12,13\}$
7	-	-	$\{1,2,3,5,6,7,8,9,11,12,13\}$
8	-	-	$\{1,2,3,5,6,7,8,9,11,12,13\}$
9	-	10	9
10	-	-	$\{1,2,3,5,6,7,8,9,10,11,12,13\}$
11	-	-	$\{1,2,3,5,6,7,8,9,11,12,13\}$
12	-	-	$\{1,2,3,5,6,7,8,9,11,12,13\}$
13	-	-	13

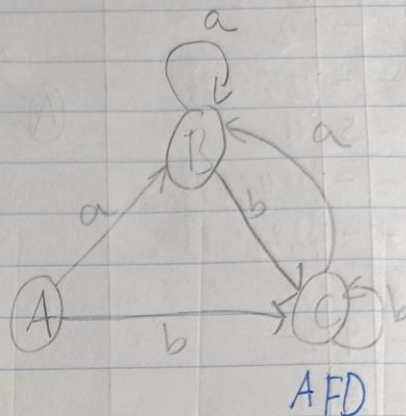


TABLA DE TRANSICIÓN DE AFD

ESTADOS	$a\epsilon^*$	$b\epsilon^*$
$\{0,1,2,3,5,6,7,8,9,11,12,13\} = A$	$\{1,2,3,4,5,6,7,8,9,11,12,13\} = B$	$\{1,2,3,5,6,7,8,9,10,11,12,13\} = C$
$\{1,2,3,4,5,6,7,8,9,11,12,13\} = B$	"	"
$\{1,2,3,5,6,7,8,9,10,11,12,13\} = C$	"	"

e). $(a/b)^*abb(a/b)^*$

